

5.3 THE QUANTITY THEORY OF MONEY (QTM)

MARKING SCHEME

1990/AL/II/14 C	1995/AL/II/18 D	2000/AL/II/22 D	2005/AL/II/21 D (77%)	2014/DSE/I/37 D (56%)
1990/AL/II/26 C	1996/AL/II/14 A	2001/AL/II/23 C	2007/AL/II/11 B (77%)	2015/DSE/I/33 A (58%)
1991/AL/II/14 A	1996/AL/II/20 C	2002/AL/II/19 C (52%)	2008/AL/II/18 D (61%)	2016/DSE/I/34 D (34%)
1991/AL/II/16 A	1997/AL/II/12 B	2002/AL/II/23 B (73%)	2009/AL/II/22 C (71%)	2017/DSE/I/36 C (64%)
1992/AL/II/20 B	1997/AL/II/22 D	2003/AL/II/11 D	2011/AL/II/24 A (59%)	2018/DSE/I/37 B (72%)
1994/AL/II/17 D	1998/AL/II/14 C	2004/AL/II/19 C (44%)	2012/AL/II/16 C (69%)	2019/DSE/I/35 A
1994/AL/II/22 D	1999/AL/II/14 D	2005/AL/II/14 C (74%)	2012/AL/II/20 B (48%)	2021/DSE/I/38 A
1994/AL/II/24 B	2000/AL/II/16 D	2005/AL/II/18 A (56%)	2013/DSE/I/32 B (74%)	2023/DSE/I/37 B
2024/DSE/II/37 D	2025/DSE/I/39 B			

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

1990/AL/II/8(a)(ii)

It all depends.

If the growth rate of the money supply approximately equals the growth rate of real output, then the inflationary consequence will be minimal.

However, if the growth rate of the money supply exceeds the growth rate of real output, inflation will occur.

1991/AL/II/2

Explain $MV = PY$ as an identity in accounting. It is only when V and Y are constant that a change in M gives rise to an equi-proportional change in P , the money price level. (M/P) is always a constant --- prediction of the classical QTM.

The classical QTM is used in conjunction with the classical theory, which states that (in the long run) flexible relative prices imply full employment. Y in this context refers to full employment output. The classical QTM requires the (long run) assumption of full employment rather than explaining full employment.

1993/AL/II/4

According to the Quantity Theory of Money, $PY = MV$

where P = price level, Y = output, M = money supply and V = velocity.

Assume V is constant. If the percentage change in income is 5% and the percentage change in the money supply is 10%, the inflation rate is (i.e. P increases by) 5% (= 10% - 5%).

(We cannot predict the rate of inflation if V is not constant.)

1999/AL/II/4

- (a) Inflation is related to monetary factors in the sense that, other things being equal, prices must rise as money supply is increased because we then have more money to spend on the same quantity of goods. [This can also be explained in terms of the quantity theory, assuming constant velocity and output.]
- (b) The quantity equation implies that the inflation rate is equal to the negative of the output growth rate when money supply and velocity are constant. In other words, prices can also rise when we use the same quantity of money to chase after a smaller quantity of goods.

2000/AL/II/8

- (a) Using the equation of exchange, we have $MV = PY$. If velocity is constant, nominal income (PY) must grow at the same rate as the money supply. As both Y and P can change, we cannot separate the impact on P and Y . If we know how Y grows, we can predict the impact of M on P .
- (b) (i) Given velocity is constant, nominal income grows at the same rate as the money supply.
 Nominal income growth rate in year 2 = $(110 - 100) / 100 = 10\%$
 Nominal income growth rate in year 3 = $(121 - 110) / 100 = 10\%$
- (ii) Inflation rate in year 2 = 10%
 Inflation rate in year 3 = $10\% - 3\% = 7\%$
- (c) If the velocity of circulation is constant and real income grows at a constant known rate, then setting the growth of money supply at a specific rate can achieve the target rate of inflation.

2002/AL/II/2(a)

True.

$$QTM: g_P = g_M + g_V - g_Y$$

$g_P > 0$ even when $g_M = 0$ as long as $g_V > g_Y$ (such as when velocity is constant and output is declining).

2007/AL/II/3(b)

The velocity of circulation of money is the number of times an average dollar changes hands – as a medium of exchange to facilitate market transactions – within a certain time period.

Using the Equation of Exchange, we can measure the velocity by the ratio of nominal GDP to money supply.

2010/AL/II/2(c)

MV represents the total amount of circulating money in the economy within a given period of time. PY or GDP represents the total money value of (final) goods and services produced in the economy within the same period. $MV = PY$ because, in a monetary economy, money changes hands whenever buyers use the money they hold (supplied by the central bank) to purchase goods and services from the sellers.

The Equation of Exchange implied that:

$$g_P = g_M + g_V - g_Y$$

which implies in turn 2 possible sources of inflation (Quantity Theory of Money as a theory of inflation):

- growth in the money supply --- i.e., $g_M > 0$, given $g_V = g_Y = 0$ (or given $g_M > g_Y$ and $g_V = 0$); and
- output decline --- i.e., $g_Y < 0$, given $g_M = g_V = 0$.

2012/DSE/II/13(d)

QTM: $MV = PY$, where M = money stock, V = velocity of circulation of money, P = general price level and Y = real output, assuming V constant. (2)

An increase in M leads to an increase in P when Y increases by a smaller proportion.

OR

An increase in M leads to an increase in P by the same percentage when Y is constant. (2)

2018/DSE/II/7(b)

$MV = PY$, where M : money supply, V : velocity of circulation of money, P : price level, and Y : output (2)

Assume V and Y are both constant. Then when M increases, P would increase by the same proportion.

OR

Assume V is constant. Then when the percentage increase in M is higher than that in Y , P would increase as well. (2)

2020/DSE/II/4C

C) M = Nominal Money Supply

V = Velocity of Money

P = General price level.

Y = Real National Income

While both V and Y are assumed to be constant in the long run. Percentage change of Money supply equal to the percentage change of price level. Continuously expansionary monetary policy (increase of M) will lead to persistent increase of general price level (increase of P) (4 marks)

2022/DSE/II/6a

(a) QTM: $MV = PY$, where (4)

M = money supply,

V = velocity of circulation of money, assumed to be constant,

P = price level, and

Y = real output.

% increase in Y being larger than % increase in M would result in a drop in P , i.e. deflation occurs.¹

2023/DSE/DSE/II/12b

(b) Quantity Theory of Money: $MV = PY$ (1)

M = money supply

V = velocity of circulation of money

P = price level

Y = real output

Assuming V to be constant; (1)

as M increases, P increases if the growth rate of M exceeds that of Y . (1)